EMG-CONTROLLED ROBOTIC ARM

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In

Mechanical Engineering

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Appendix B

2026 – 2027

CERTIFICATE

This is to certify that the B.E. project entitled "**Emg-controlled robotic arm** " is a bonafide work of "**Siddhant Dalvi**" (EU1235009), "**Nivan Gharat**" (EU2235010), “**Raj Mishra**" (EU1235005) and "**Narendra Pawar**" (EU1235010) submitted to University of Mumbai in partial fulfilment of the requirement for the award of the degree of "Bachelor of Technology" in "Mechanical Engineering" during the academic year 2026-2027.

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Abstract

This project focuses on the development of an EMG-controlled robotic arm. The proposed system is intended to translate muscle activity into control commands for a multi-finger prosthetic mechanism. Surface electromyography (EMG) is considered as the user input, an ESP32-based controller is planned for signal processing and command generation, and five MG996R servo motors are used as the finger actuators.

A tendon-driven hand built using 3D printing provides the base for moving fingers and grabbing objects. The goal of this project is to bring mechanical design, embedded control systems and biosignal-based interaction into a single compact prototype. Work on the design includes picking the setup for the actuators making the 3D-printed parts planning how the tendons will run through the hand mounting the servos and creating a control sequence that turns EMG signals into finger movement. Far the project has completed 3D printing choosing the right components putting the mechanical parts together testing the servos and checking the EMG sensors. The next steps are to install the EMG sensor connect the controller and the actuators get the fingers to move together smoothly test gripping and do a full evaluation with measurable results. These final parts will need to be completed and written down.

The report follows the structure as the sample report given: introduction, literature review, requirement analysis, design and development fabrication and control, testing and discussion and conclusion, with future work.

Keywords: hand, EMG, ESP32 MG996R servo, tendon-driven mechanism, adaptive grip, 3D printing.

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List of Abbreviations

Sr. No.	Abbreviation	Full Form
1	EMG	Electromyography
2	CAD	Computer-Aided Design
3	CAM	Computer-Aided Manufacturing
4	MCU	Microcontroller Unit
5	CPU	Central Processing Unit
6	GPIO	General-Purpose Input/Output
7	ADC	Analog-to-Digital Converter
8	PWM	Pulse Width Modulation
9	IDE	Integrated Development Environment
10	USB	Universal Serial Bus
11	DC	Direct Current
12	AC	Alternating Current
13	V	Volt
14	mA	Milliampere
15	FSR	Force-Sensitive Resistor
16	ESP32	ESP 32-bit Microcontroller Series
17	3D	Three-Dimensional
18	DOF	Degrees of Freedom
19	RAM	Random Access Memory
20	ROM	Read-Only Memory
21	PCB	Printed Circuit Board
22	I/O	Input/Output
23	STL	Standard Tessellation Language

CHAPTER 1 INTRODUCTION

i. MOTIVATION

The human hand does things by moving the fingers, thumb and wrist together. If the hand doesn't work well it can make tasks hard take away independence and make life less enjoyable. A prosthetic hand tries to help by giving a replacement that the user can control.

Traditional prosthetic hands can be for looks or work using the bodys movements or electricity. A powered hand can use parts to move fingers one by one or in groups. But a good design has to balance things like how well it works how complicated it is, how heavy it is, how much power it needs how much it costs, how easy it is to fix and how simple it is to use.

Electromyography or EMG looks at the electricity made when muscles work. Small sensors, on the skin can pick up this signal without hurting the person. This signal can be used to tell a machine what to do. In this project muscle activity is meant to be the users command while an ESP32 controller and servo motors move the fingers of a 3D-printed hand that uses tendons.

The goal is to create an cheap model that shows how biology, electronics and movement connect. This model is not meant to be a medical device. It is meant to help learn the engineering ideas needed for a hand that moves with EMG signals and has multiple finger

2. PROBLEM STATEMENT

A hand mechanism that is operated or fixed in motion does not respond naturally to a user's muscle activity. At the time an EMG-controlled multi-finger hand must deal with changes in placement against the skin, muscle fatigue, electrical noise, the load, on the actuator and mechanical friction. If these issues are not considered the hand mechanism's output can become unstable delayed or not consistent.

The engineering challenge is to create a hand mechanism in which EMG activity can be sensed and converted into repeated movement commands while the fingers are moved by servos through a tendon-driven system. The hand mechanism should provide a clear control process allow for calibration and make it possible to test grip actions. The project must also divide functions that have already been tested separately from functions that are still not confirmed after they are put together..

3. OBJECTIVES

Specific objectives are:

- To create a mechanical hand structure composed of articulated fingers and tendon actuated mechanism.
- To create hand parts by 3D printing and assemble the mechanical structure.
- To choose and attach 5 servo motors of type MG996R to the fingers in the final product.
- mechanical arrangement.
- To gather muscle activity input from an EMG sensor and test the signal when the muscle is at rest and contracting.
- To implement the desired communication between EMG signal and actuators using an ESP32 based controller.
- To devise an approach for calibrating and controlling the mapping of muscle activity to finger movements.
- To document objective data and symptoms of limitations and potential improvement

4.SCOPE

- The project contains design, 3D printing, tendon routing, servo design, EMG sensing, embedded control and prototype level testing. The main goal of the project is to demonstrate a control chain from muscle activity to mechanical finger motion.

- The project depends on the finger geometry for simple grip tasks. For example the project can use a power grasp around an object and a lighter grasp of a suitable test object. These are test targets, not claims that the integrated prototype has already passed the tests.

- The scope of the project does not include trials, certification for medical use long term daily use testing, advanced pattern recognition, with a large EMG dataset and a complete commercial socket and wearable fitting system. Measurement and validation are required for any force, endurance or usability claim.

5.Summary

The project involves an integration of mechanical engineering, embedded control and biosignal sensing. The primary technical difficulty is the movement both the fingers need and the fact that a movement is supposed to be reproducible and controllable based on the EMG signal. The next chapters set the background and requirements for this development.

CHAPTER 2
REVIEW OF LITERATURE

2.1 LITERATURE REVIEWS

2.1 Prosthetic Hand Systems

Prosthetic hands are made to be used to replace part of the functions of a missing hand. Their architecture is based on the task they are to perform, the control inputs available, the type of actuator used, as well as the size/weight constraints. Passive hands can give appearance and positioning but they cannot grasp. Body powered designs are those in which a cable or harness is moved to control a terminal device. Externally powered hands, electric actuators and control system – may enable more than one grip configuration.

In evaluating the mechanical performance of upper-limb prostheses, the reviews note the importance of grasping ability, force, mobility, speed, weight, reliability, and user control. Belter et al. examine mechanical design-performance specifications link in anthropomorphic prosthetic hands. This reinforces the requirement for an explicit evaluation of the proposed hand, rather than just relying on what looks good.

The voltage change of muscle activation is measured with electrodes attached to the skin surface, called surface electromyography. Electrode position, skin preparation, contact resistance, muscle activation, motion artifacts and electrical interference are factors that influence the signal. De Luca describes how surface EMG can be used in biomechanics and discusses factors affecting the interpretation of the signal. When an EMG signal is used to trigger or modulate an action of a mechanical system, these issues have significance.

In the case of a simple prototype, the conditioning of the EMG signal can be performed by the sensor module and the sampling can be done by the microcontroller. Threshold-based method can be used to distinguish low activity and contraction state after calibration. More sophisticated methods can be used to identify features across a time window and recognize multiple gestures. The correct way to do it is a function of the signal quality, the number of desired commands and the amount of computing resources.

2. Embedded Control Using ESP32

The ESP32 family is a microcontroller family that offers general purpose I/O, analogue input on compatible pins and timer-based control options. The pin mapping, ADC characteristics and servo-control type is specific to the ESP32 board and software used.

The controller in the proposed system is supposed to collect the conditioned output of the EMG, filter/smooth the collected samples, use a calibrated decision rule and output position commands to the servos. The controller should NOT directly drive the servo motors. A power source is needed outside the servos to power them, and a common signal reference, if specified by the circuit design. This will require the power supply to be sized for the current that it will draw when multiple servos are moving or stalled..

3. Tendon-Driven Finger Mechanisms

The tendon driven mechanism is a system that moves force from a motor to a finger joint by using a tendon that goes around the finger. This setup helps make the fingers lighter. It lets the motors stay in the palm or forearm of in the fingers themselves.

You must think about things when building this: friction in the tendon how the tendon is routed where it's attached how it returns and if there's any slack. The finger might need parts, like springs or another tendon to help it open after bending.

The moment arm. The distance that creates turning force. Changes as the tendon path moves. So the finger's motion isn't a simple copy of how much the motor turns. You can change how the finger closes by moving where the tendon is attached or by adjusting the tension in the tendon.

Because of this it's important to record the tendon route. Use. Cad drawings to keep track. Any changes must come from testing and steps. Don't assume that the tendon path is the same, on every finger. Each one may need its setup**.**

4. Servo Actuation

A hobby servo will almost always have a motor, gears, a way to know where it is and a controller inside. A signal or control message tells the servo where the arm needs to go. The MG996R series is used in hobby robots because it has gears on the output. The power, the speed, how electricity it uses and how far it can move all change depending on the model and how it is being used. You must look at the sheet or the sellers details for the specific one you bought when writing the final report.

The maximum torque a servo can handle is not the power that is always being used. When it is stuck or almost stuck it uses electricity gets hotter and wears out faster. Much tightness, in the string can make the servo make noise or get too hot or damage the parts in the hand. So using stops to limit movement being careful with how it moves and having the right power source are really important.

2.5 Comparison of Existing Approaches

Approach	Typical input	Strengths	Limitations relevant to this project
Passive prosthesis	No active control	Simple, light, low maintenance	Cannot actively close fingers
Body-powered device	Harness/cable movement	Direct mechanical feedback; no battery for actuation	Harness movement may be demanding; limited grip options
Externally powered hand	Switch, EMG or other input	Can produce powered grasping	Requires actuators, battery and control electronics
Threshold EMG prototype	Muscle activity above calibrated threshold	Relatively simple to implement and explain	Sensitive to electrode placement and signal variation
Pattern-recognition EMG	Features from multiple channels or time windows	Potentially supports multiple commands	Needs data collection, processing and robust validation
Proposed project	EMG sensor + ESP32 + servo- actuated printed hand	Affordable educational platform for mechanical/control integration	Integrated control and performance still require validation

CHAPTER 3 REQUIREMENT ANALYSIS

3.HARDWARE REQUIREMENTS.

Building the Emg-controlled arm needs a mix of mechanical, electrical and electronic parts.
The list of requirements tells you what mechanical, electronic and control tasks the prototype must perform.

1. MECHANICAL COMPONENTS

All moving parts should be protected from sharp edges and pinch points during benchtesting.
Servo mounts should be able to move under the tendon load without binding or excessive interference and should be adjustable or replaceable.
Design must make it easy to reach fasteners and tendons and electrical components for maintenance work. The design needs to give access to fasteners and tendons and electrical parts so that maintenance can be done properly. Every part that needs checking or fixing should be available without difficulty. The design should ensure that fasteners and tendons and electrical parts are all easy to get to when needed. Access to fasteners and tendons and electrical parts is important for maintenance tasks. The design should not hide fasteners and tendons and electrical parts where they cannot be reached. Maintenance workers need to be able to access fasteners and tendons and electrical parts easily. The design must allow for access to fasteners and tendons and electrical parts, during maintenance. All fasteners and tendons and electrical parts should be placed where they can be reached during maintenance.

3.2 Software Requirements

- The control program is to start the sensor input and the servo outputs and to take baseline readings, smooth/filter the input as necessary, compare the processed signal to calibrated thresholds, and control the servos to a defined safe position. The program should have an open/release state explicitly defined, and should not oscillate quickly when the signal drops close to a threshold.
- Sensor sampling and baseline measurement.
- Noise reduction or smoothing appropriate to the sensor output.
- Calibration of activation and release thresholds.
- Servo-angle limits and a controlled movement rate where supported.
- Clear separation between input processing, grip-state selection and servo output.
- Serial or other diagnostic output during development to record signal values and state transitions.
- ESP32 board support, to write and upload the code.
- ESP32 servo library (such as ESP32Servo) for PWM control of five servos.
- Serial Monitor / Serial Plotter, to view raw and smoothed EMG values while setting thresholds.
- CAD software: SolidWorks or Creo for the hand model, .
- Slicer (such as Cura) for the 3D prints**.3333**

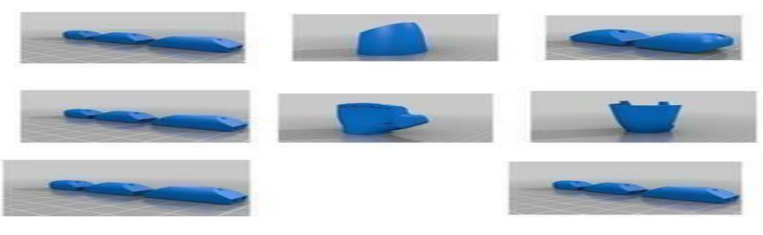


FIG 3.2 3-D diagram of parts

3.3 HARDWARE REQUIREMENT

- 3D-printed structure: palm, thumb, four fingers, joints and an open forearm shell (PLA).
- Microcontroller: ESP32 development board. It has an ADC for the sensor, PWM pins for five servos and built-in Bluetooth/Wi-Fi. Its logic level is 3.3 V.
- EMG sensor module with electrodes: gives an analogue output you read on an ESP32 ADC pin. Confirm the exact model and its supply voltage.
- Servos: 5 × MG996R, one per finger. Datasheets quote about 4.8 to 7.2 V and roughly 10 kg·cm stall torque, so check your units.
- Tendons and return: low-stretch fishing line, with elastic band or spring to open the fingers.
- Power supply: a separate regulated 5-6 V supply for the servos, enough current for the combined power of the servos plus a few amps for safety, as per the datasheet. Share a common ground with the ESP32, and add a power switch.
- Wiring: breadboard and jumper wires for now, with a soldered board or PCB for the final version



FIG 3.3 ROBOTIC ARM MODEL

CHAPTER 4
DESIGN AND DEVELOPMENT

1.INTRODUCTION

The design and development stage is about to turn the project requirements into a functional robotic mechanism.This system is composed of mechanical structure, electronic sensors, microcontroller, servo motors and embedded software.The mechanical design gives the structure for moving, the electronics system detects the muscle activity and changes the signal into control commands. Development process includes conceptual design, CAD modelling, Actuator selection, fabrication, Assembly, electrical integration and programming.The design should allow easy access to electronic components, and allow the mechanical system to be inspected and modified during the tests.

2.DESIGN OBJECTIVES

Some of the principal design goals are:
❑Make the robotic arm mechanically functional.
❑Identify appropriate mounting places for the actuators.
❑Integrate an EMG sensing system.
❑Make the signals processable by the microcontroller.
❑Allow controlled movement of the servos.
❑Ensure proper mechanical alignment.
❑Minimize interference between moving components.
❑Provide secure electrical connections.
❑Allow easy testing and maintenance.
❑Support future improvements.

4.3 CONCEPTUAL AND MECHANICAL DESIGN

The conceptual design starts with the identification of the movements to be performed by the robotic arm.The mechanical structure is then subdivided into appropriate parts including base, arm segments, forearm, palm and fingers.The amount of arm joints will vary based on the desired motion and the end configuration of the prototype. All joints should have the necessary freedom in motion, but have adequate mechanical support at the same time.A robotic hand (if used) is designed to aid in the opening and closing of fingers, and gripping objects.Servo rotation can be translated to finger bending by a tendon driven mechanism.Servo rotation can be transmitted to bending the fingers by a tendon driven mechanism. The placement of the actuator and routing of the tendons should be designed to minimise friction and interference. The last mechanical sketch should include an inspection for:

- Joint alignment.
- Structural strength.
- Actuator mounting.
- Tendon routing.
- Movement range.
- Mechanical clearance.

.

4.4 MAIN STRUCTURE DESIGN

The arm arm mechanism and its movable parts is supported by the main structure.The structure should be sufficiently rigid to avoid too much deflection when it is working.The mass of the servos, wiring and mechanical components must be considered in the design. Holes and brackets to be installed to ensure secure mounting.In the case of an arm with a base, the base must be sufficiently stable against the forces that are created during movement. These final dimensions should be based on the CAD model and fabrication requirements.

4.5 SERVO-MOTOR MOUNTING ARRANGEMENT

The positioning of the servo motor is essential for the proper transfer of the motor's motion to the mechanical system and must be securely made.The mounting brackets should stop unwanted movement during loading of the actuators.In case of a tendon-driven finger mechanism the servo horn or the servo spool should pull the tendon along an appropriate path. The mounting system should allow for adequate clearance for the motor body, wiring, and moving parts.The chosen servo position must also be easily adjustable and replaceable if needed.

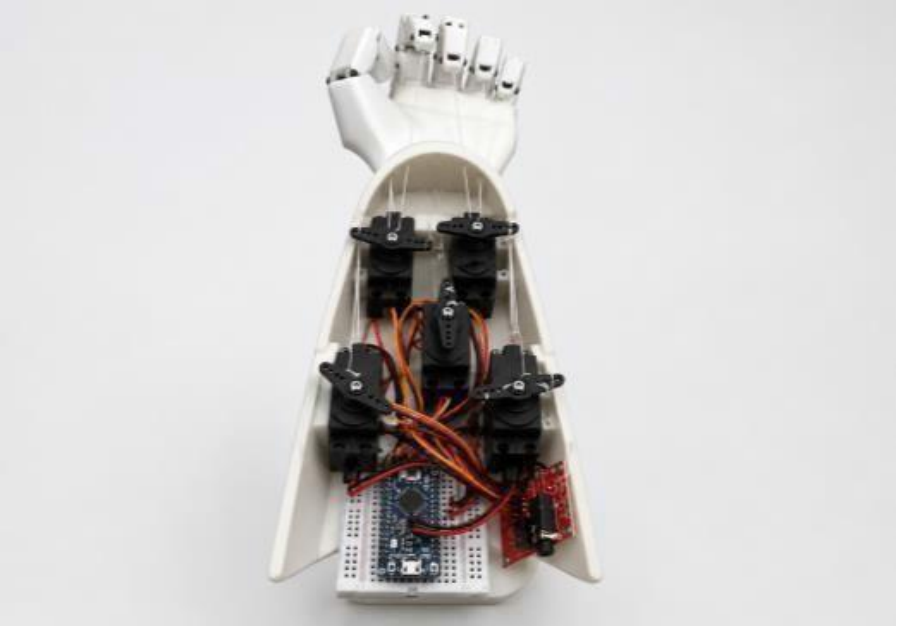


Figure 4.5: Motor Mounting Arrangement

4.6 ACTUATION SYSTEM

The actuation system is responsible for turning the electrical control commands into mechanical movement.The microcontroller generates signals to control the servo motors

.The output of a servo, when it moves, can transmit motion to a mechanical joint or to a tendon attached to a finger.The position of the actuators is controlled by the control program.

Mechanical limits of the structure should not be exceeded by the motors, so the actuation system should be properly set.The choice of actuator and means of transmission should depend upon the load and required movement.

4.7 EMG-Based Control System

TThe EMG control system is a system that uses muscle activity as a signal and produces a control command based on the muscle signal.The sensor is attached to a proper input of microcontroller.During operation, controller reads the value of a sensor and compares it with the parameters calibrated.When the necessary condition for activation occurs, the control algorithm generates a motion command.Servo Control Subsystem Control command sent.This process is stable and performs well if the signal stability, calibration, control logic and actuator response are stable.

4.8.CAD MODELLING AND ASSEMBLY

- The three-dimensional geometry of the robotic arm is made using CAD modelling. The modelling process consists of the following steps:
- Defining the dimensions of the overall model.
- Making the base and structural elements.Modelling the joints.
- Designing the palm and fingers.
- Creating servo mounting brackets.
- Providing space for the tendon mechanism.
- Checking component clearances.
- Assembling the parts in the CAD environment.
- Reviewing the movement and interference conditions.
- Preparing the models for fabrication.

4.09 COMPONENT SELECTION

The components selected are dependent on mechanical, electrical and control needs..

COMPONENTS	PURPOSE
EMG SENSOR	Detects muscle-related electrical activity
Microcontroller	Processes sensor readings and generates control commands
Servo Motors	Operate mechanical joints or fingers
3-D Printed Structure	Provides the mechanical framework
Tendon Mechanism	Transfers actuator movement to the fingers
Power Supply	Provides electrical energy to the system
CAD Software	Supports mechanical design and assembly

Table 4.09 Component Selection

4.10 SUMMARY

In this chapter the conceptual design, mechanical structure, actuator arrangement, EMG control system, and the selection of components are described. The design involves a combination of mechanical and electronic control, with the aim of creating a robot mechanism that is able to move in response to muscle movements. Structural components are customized with the use of CAD modelling and 3D printing.Fabrication, assembly, electrical integration and programming are all next steps in the process.

CHAPTER 5
FABRICATION, ELECTRICAL AND CONTROL SYSTEM

1. INTRODUCTION

The present investigation focuses on the practical development of the EMG-Controlled Robotic Arm.The main activities include mechanical fabrication, actuator installation, sensor interfacing, circuit assembly, microcontroller programming, and system integration.The mechanical and electronic subsystems must be checked individually before integrated testing.This approach helps identify assembly errors, wiring problems, and software issues before evaluating the complete system.

2. FABRICATION PROCESS

The fabrication process begins with preparing the mechanical design and identifying the parts
that need to be manufactured.The CAD models are converted into printable files using slicing software.The parts are then produced using a 3D printer.After printing, the components are inspected for dimensional accuracy, surface defects, and compatibility with the intended assembly.

The fabricated parts are assembled using appropriate fasteners and mounting arrangements.Any parts that do not fit correctly are adjusted or redesigned before final assembly.

3.MAIN FRAME FABRICATION

:

The main structure is fabricated according to the CAD design.
The printed components are checked to ensure that they have the required dimensions and mounting features.
The structural parts are assembled in the intended arrangement.
During assembly, attention is given to joint alignment, rigidity, and clearance between moving components.
The structure should be inspected for loose connections and areas that may interfere with movement.

4. COMPONENT MOUNTING AND ASSEMBLY

Mechanical components are assembled following the chosen design.Servo motors are mounted in their respective places.If a tendon mechanism is employed, it is attached to the proper moving fingers. The position of the EMG sensor and microcontroller ensures the wiring stays in place and doesn't block the mechanical motion. The assembly should be designed to make the control electronics easy to reach to aid in debugging and maintenance..

5. SERVO-MOTOR INSTALLATION

. The servo motors are set into place and fastened with appropriate mounting systems.Servos are tested separately prior to tying them into the full mechanical linkage.Usable operating range and direction of rotation is checked. For tendon-driven fingers the tendon attachment is fine-tuned to achieve appropriate finger motion.Actuator positions should be set up so that there is not too much tension or mechanical interference.It's important to check the installed servos for suitability under the mechanical load.

6. EMG SENSOR installation

The EMG sensor is wired as stated in the manufacturer's wiring diagram.The placement of the electrode should be appropriate for detecting the activity from the muscle chosen.The output of the sensor is attached to the compatible microcontroller input. On initial testing, the output is tested to ensure that it changes when the muscle is relaxed and contracted. The signal readings are monitored to check if calibration and filtering is needed.The sensor must be mounted and the cables must not interfere with the working of the robot mechanism.

7. Electrical System Assembly

Electrical System Assembly is the connection of the EMG sensor, the microcontroller, the servo motors and optional force sensors.Wiring needs to meet the electrical needs of the chosen components The power supply should supply the voltage that is required and the current required by the servo motors.Before applying power, check the connections between the microcontroller and the sensors.Follow for the electrical assembly: check for loose wiring, wrong polarity, short circuits and inappropriate power connections.

5.8 ELECTRICAL CIRCUIT DESIGN

Muscle activity detected by the EMG sensor gives an output that corresponds to the activity.The output is fed into the microcontroller which runs the control algorithm.Control signals are sent to the servo motors which generate the desired mechanical action.The actual ground, power supply, servo signal pin connections, sensor pin connections, and controller must be included on the actual circuit diagram.

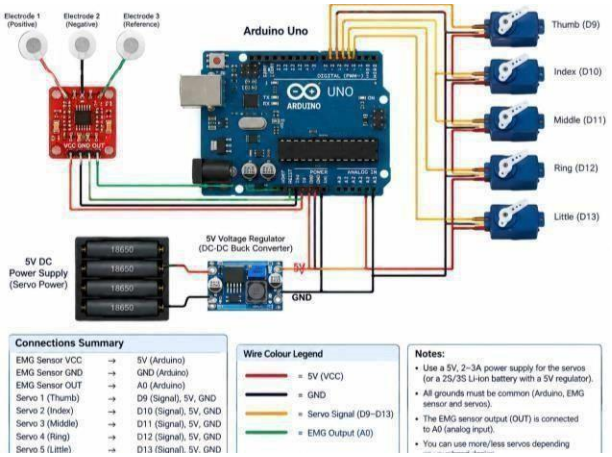


Figure 5.10: Electrical Circuit Diagram

5.9 CIRCUIT CHECKING AND TESTING

The electrical connections are individually tested prior to installation of the full system. The following must be checked:

•correct connections to sensor(s)

•correct pin assignment for microcontroller(s)

•supply voltage (v) is adequate

•correct input connection to servo(s) signal(s)

• correct grounding, including adequate grounding of the sensor(s) and the power supply(s)

• Secure electrical connections (no broken or frayed wires)

The system should first be tested while the mechanical load is disconnected or otherwise safely supported, if possible.After the individual checks, the electronic components can be integrated with the mechanical assembly.

10. CONTROL SYSTEM

The control system is developed to convert the EMG signal into a movement command.The program reads the sensor output, and compares the measured value to a calibrated threshold.Once the activation condition is fulfilled, the controller produces the necessary servo commands.The limits in the control program should be set to prevent the servos moving beyond any mechanical safe limits.The final control logic is dependent upon whether the system is based on the simple activation detection or a more in-depth and advanced process of choosing various movements.

11. CONTROL PROGRAM

The general control program flow is:

- Start the system.
- Initialize the microcontroller.
- Configure the sensor input.
- Initialize the servo motors.
- Read the EMG sensor.
- Process the sensor output.
- Compare the reading with the calibrated threshold.
- Determine the required movement.
- Generate the servo command.
- Check optional sensor inputs.
- Repeat the process.

5.12 MOTOR TESTING

The servo motors are tested individually before integrated operation. The testing procedure should verify:

- Correct direction of rotation.
- Appropriate movement range.
- Repeatability.
- Mechanical alignment.
- Response to control signals.
- Stability under the intended load.

For tendon-driven fingers, the motor movement should be checked against the resulting finger movement.

5.13 FINAL ASSEMBLY CHECK

After the mechanical and electrical subsystems have been assembled, the complete system is inspected.

The inspection should confirm that:

- The mechanical components are securely mounted.
- The servo motors are correctly positioned.
- The tendon mechanism operates freely.
- The EMG sensor is connected correctly.
- The control circuit is properly wired.z
- The power supply is suitable.
- The program is loaded onto the microcontroller.

The moving components do not interfere with cables or other parts.



Fig-5.1.1 Fabrication and assembly process

5.14 SUMMARY

This chapter describes the fabrication, assembly, electrical integration, and control-system development of the robotic arm.The mechanical components and electronic subsystems must be
checked individually before complete operation.The final implementation details should be updated according to the actual hardware and software used in the prototype.

CHAPTER 6

TESTING, RESULTS AND DISCUSSION

1.INTRODUCTION

Testing is required to evaluate whether the EMG-Controlled Robotic Arm operates according to its intended design.The purpose of testing is to verify the performance of individual components and determine whether the complete system can translate muscle activity into mechanical movement.The testing process includes sensor verification, servo testing,

mechanical inspection, control-program testing, and integrated-system evaluation.

2.OBJECTIVES OF TESTING

To verify operation of the EMG sensor, observe the change in the output of the sensor during the muscle contraction, verify the acquisition of the signals by the microcontroller, test the movement of the servo motor, verify the mechanical assembly, and check the response of the control program, evaluate relationship between muscle contraction and robotic movement, identify errors in mechanical/electrical systems, investigate the performance of optional force sensors, and identify possible improvements.

6.3 TESTING METHODOLOGY

Testing should be done in a phased manner. Stage 1: Mechanical Inspection, check structure, joints, fasteners, actuator mounts and tendon mechanism. Stage 2: Electrical Inspection: Check sensor connections, power supply, grounding and servo wiring. Stage 3: Sensor Testing: Note the EMG output from the sensor when muscle is relaxed and when muscle is contracted. Stage 4: Actuator Testing Run a test program on the servo motors and check their motion. Stage 5: Control Program Testing See if the program works as expected when the EMG sensor output is changing. Stage 6: Integrated-System Testing: Test the assembled system and monitor the signal to movement sequence. Stage 7: Performance Evaluation Document results, pinpoint areas for improvement, and determine competencies that have been met.

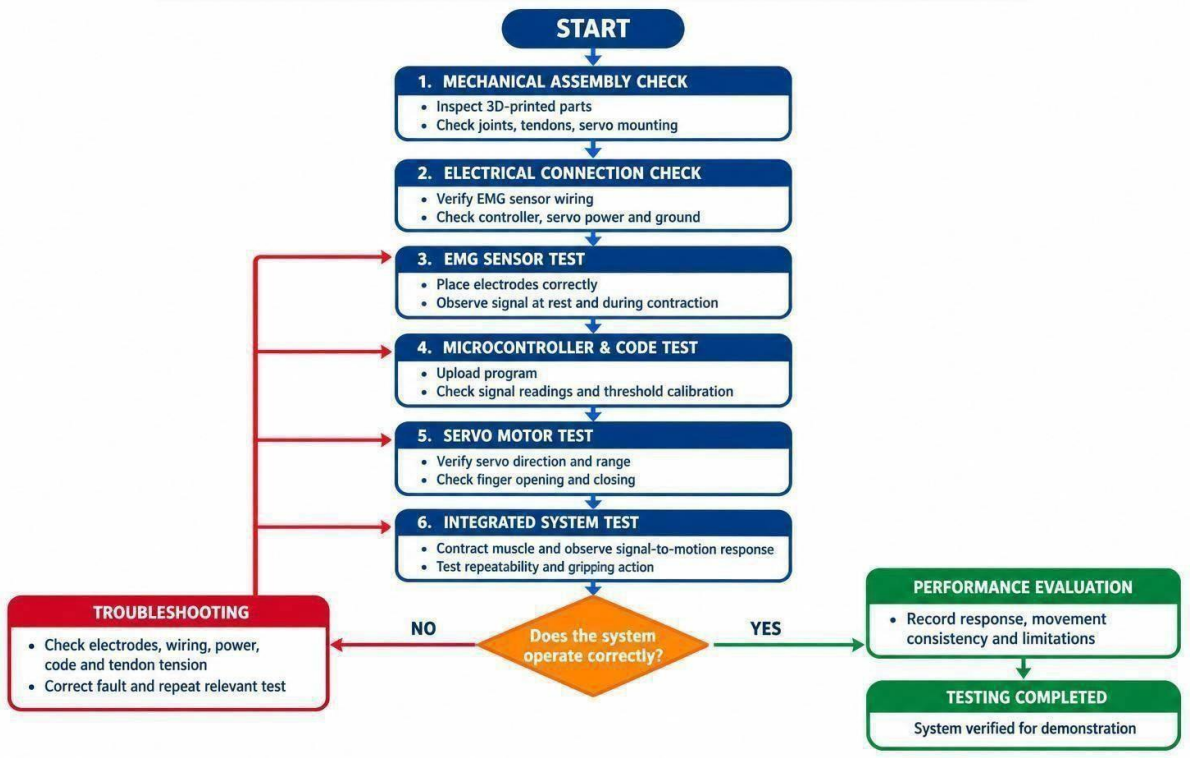


Figure 6.3: Testing Methodology Flowchart

4.ASSEMBLY CHECKING

Prior to turning on the power the mechanical assembly must be inspected.The inspection should

Work on structural alignment, mounting security, joint clearance and positioning of actuators. The mechanism for the tendon routing and tension should be checked.Loose

Any fasteners, mechanical obstructions should be rectified prior to testing.

5.ELECTRICAL CIRCUIT CHECKING

Check electrical connections against circuit diagram.The sensor,

The microcontroller, servos and power supply should be wired in the same way as the specification prescribes.

Before operating the supply voltage should be measured.

Loose connections, polarity problems and potential short circuits should be checked in the wiring.

circuits.

6.EMG SENSOR testing

It is required to observe the output of EMG sensor under different muscle conditions to test it. First, measure the output from the sensor as the muscle chosen is relaxed.

Then record the output when the muscle is contracted in a controlled manner. Repeat measurements to see if there is a difference between the relaxed and active states is consistent.

The observations can be recorded in the following table.

Trial	Muscle condition	Observation
1	Relaxed	Low muscle activity
2	Contracted	Increased muscle activity
3	Relaxed	Muscle activity returned to a low level
4	Contracted	High muscle activity detected
5	Contracted	Muscle contraction maintained

7. SERVO MOTOR TESTING

A test program should be used to operate each servo motor.
The test should check the direction of rotation, operating range and reaction to control.
signals.
The motor should then be subjected to the mechanical load it is expected to be used for. The main
During the motor testing the following observations were made: Motor starts properly.
Motor rotates correctly.7
Motor responds to control input.
No abnormal connection problem was noticed.
Issues identified and corrected to get required motor operation.

8. CONTROL PROGRAM TESTING

The control program needs to be tested to ensure that it reads the EMG sensor and produces
The commands the actuator was supposed to execute.The desired actuator commands.The test should involve relaxed-muscle conditions, active-, and passive-conditions.
Repeated activation, muscle conditions. The program should be screened for unwanted
Unstable switching, wrong servo commands and triggers.If the measured signal vibrates about the activation level, calibration, filtering or hysteresis may be necessary.

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6.9 CODE DEBUGGING

When testing a program during coding, it should be identified and corrected if any errors or wrong answers are found.
When debugging, one can check:
•Sensor input readings.
•Microcontroller pin assignments.
•Servo initialization.
•Threshold values.
•Control logic.
•Servo position limits.
•Timing and repeated commands.
•Electrical communication.
Once the program has been altered, the program should be retest again.
The process should be repeated until the desired control behaviour is accomplished or Limits that still exist are identified

6.10 EMG-CONTROL MOVEMENT TESTING

The integrated system needs to be tested and need to find out if muscle activation makes it produce what.
intended robotic movement.
The first step of the test is to verify the signal from the EMG sensor.
A control program is then employed to detect the activation of muscles and to produce the necessary
servo commands.
This mechanical motion is then observed. These parameters can be recorded:
•Sensor activation consistency.
•Correctness of the movement command.
•Movement response time.
•Trials that were successful.
•Number of false activations.
•Repeatability of movement.

6.11 MECHANICAL MOVEMENT TESTING

It is important that the mechanical assembly be moved as it would in the desired application. During the inspection, any interference between moving parts should be identified.
Joint friction, structural movement, tendon tension, and actuator should be investigated as well.
alignment.
If the hand is attached to the robot, make sure the fingers move up and down.

12. INTEGRATED-SYSTEM TESTING

Integrated testing is the assessment of the entire control chain.

It is being operated by toning the selected muscle and watching the output of the sensor, and the output is being sent to a monitoring device.The selected muscle is being toned and the output is being monitored with the sensor output being sent to a monitoring device.

The servos respond to the microcontroller's commands and the mechanical output. The test should identify if the system is producing the desired motion over time.

The following sequence should be checked:

Muscle Activation → EMG Signal Detection → Microcontroller Processing → Servo Movement → Robotic Arm Response

13. TESTING OBSERVATIONS

Observations during the test were taken for various subsystems. The

The observations are used to determine if each subsystem was ready for complete system operations.

SR. NO.	TEST	PURPOSE	OBSERVATION/RESULT
1	Mechanical assembly	Check mechanical assembly	Proper assembly verified
2	Electrical checking	Check electrical connections	Connections checked
3	EMG sensor testing	Verify muscle related signals	Circuit response checked
4	Servo motor testing	Verify actuators operations	Motor operation verified

5	Program Testing	Check control logic	Program tested
6	Integrated Testing	Evaluate signal to movement response	The system was tested for robotic hand movement in response to EMG signals.
7	Mechanical Movement	Check movement and interference	The moving components were checked for smooth movement and interference.

Table 6.1: Testing Summary

14.TEST RESULTS

The testing activities helped to verify the major functions of the robotic arm.
The mechanical assembly was tested prior to electrical testing. Design, wiring and testing of the electrical circuit. Motor testing was conducted to ensure the running of the motor and the rotation.
Programs for controlling the pests were developed and tested. The necessary debugging was carried out during the code testing.

15.RESULTS AND DISCUSSION

The results should be analysed to determine whether the robotic arm meets its intended objectives.
The EMG sensor measurements can be used to evaluate the difference between relaxed and active muscle conditions.
The servo tests can be used to assess if the actuators are functioning correctly in response
to the
generated commands.
Combined system tests can be performed to check if EMG is able to be translated into mechanical movement.
Possible causes of unstable response: Signal noise, inappropriate thresholds, unstable power supply, mechanical friction, or faulty control logic.
The discussion should be linked to the project aims and highlight any continuing constraints.

16.PROBLEMS IDENTIFIED DURING TESTING

Some possible issues that could be explored are:
•Unstable EMG readings.

- Difficulty separating muscle activation from background noise.
- Incorrect activation thresholds.
- Servo movement in the wrong direction.
- Excessive actuator loading.
- Loose mechanical connections.
- Tendon slippage.
- Friction at finger joints.
- Inconsistent finger movement.
- Voltage drops during servo operation.
- Unwanted servo activation.
- Mechanical interference.

All the noted problems were addressed if needed and tested prior to final testing

6.17 PERFORMANCE EVALUATION

The individual and integrated tests should be used to evaluate the performance of the robotic arm.

The assessment may take into account the following:

- Signal detection.
- Control reliability.
- Actuator response.
- Mechanical stability.
- Movement repeatability.
- Gripping performance.
- Electrical reliability.
- Ease of calibration.

As far as possible, numerical values should be utilized.

For instance, the number of accurate control commands can be used to determine control accuracy.

Total number of valid test commands divided by number of commands.

Response time is the time between an activation event (noted) and the associated mechanical response.

It is important that the measurement method is clearly stated in the report.

6.18 LIMITATIONS OF PRESENT SYSTEM

Potential limitations include:

- Variability in muscle-generated electrical signals.
- Sensitivity to electrode placement.
- Need for individual signal calibration.
- Mechanical limitations of the printed structure.
- Finite servo torque.
- Friction in the tendon mechanism.
- Limited movement accuracy.

- Difficulty distinguishing multiple commands using a single threshold.
- Power-supply constraints.

6.19 FUTURE IMPROVEMENT BASED ON TESTING

- Improving EMG signal filtering.
- Refining threshold calibration.
- Adding hysteresis to reduce unwanted switching.
- Improving servo mounting.
- Optimizing tendon routing.
- Modifying finger-joint geometry.
- Improving the structural strength of printed components.
- Adding calibrated fingertip force sensors.
- Developing more reliable grip-selection logic.
- Introducing wireless data logging.
- Investigating advanced EMG classification.

6.20 SUMMARY

It is the author's intention to share in this chapter the methodology for evaluating the mechanical, electronic and

Control subsystems for EMG Controlled Robotic Arm.

The tests are designed to determine if the system is capable of detecting muscle activity and processing it,

Produce necessary mechanical action, the signal.

All final results should be calculated based on actual observations and measurements made.

from the prototype.

CHAPTER 7
CONCLUSION AND FUTURE WORK

1.CONCLUSION

The EMG-Controlled Robotic Arm project is about creating a robotic arm that can be controlled by the electrical signals generated when humans move their arms.The proposed system utilizes an EMG sensor, microcontroller, servo motors and a mechanical structure to translate muscle movements into mechanical motion.3D printing can help create custom mechanical parts and the microcontroller offers a programmable platform for signal processing and control of actuators.A tendon-driven finger mechanism can be employed to convert the servo motion to bending motion of the fingers and gripping motion of the object. The project also serves as a base for studying different gripping configuration and fingertip force feedback. Mechanical design, electronic integration, embedded programming and hands-on testing are all part of the development process. The final performance of the system should be determined by experimental testing and evaluation of the EMG sensor, the actuators, the mechanical assembly, and the integrated control system. Overall, the project is an example of human-machine interaction in the form of muscle-driven robot motion. It also offers scope for further study of robotics, embedded systems and assistive technology.

2.FUTURE WORK

The following are some of the changes that may be sought for further improvement:

- Improvements to EMG Processing: Implement more robust and repeatable methods for signal filtering and calibration, to avoid unintended muscle activation.
- Advanced Movement Recognition: Develop more sophisticated methods for movement recognition between different muscle activation patterns.
- Improvements to Mechanical Design: Optimize the structure of the arm, finger joints, and mounting of actuators.
- Improvements to Tendon Transmission: Reduce friction and increase the repeatability of the fingers' movements and investigate how grip adjustments can allow for improved repeatability.
- Add Force feedback: Add calibrated fingertip sensors that can measure contact and explore the relationship between changing the grip and repeatability.

- Additional Degrees of Freedom: Add independently controlled joints if more complex robotic-arm movement is required.
- Wireless Monitoring: Use compatible wireless hardware to transmit selected sensor readings and system-status information.
- Data Logging: Record EMG measurements and movement commands to evaluate performance.
- Improved Power Management: Optimize the electrical supply for stable operation under actuator loading.
- Advanced Control Algorithms: Investigate pattern recognition and machine-learning techniques after sufficient experimental data has been collected.
- Safety Improvements: Add appropriate movement limits, fault handling, and electrical and mechanical protections.

7.3 FINAL REMARK

The EMG-Controlled Robotic Arm is a project that brings together design, electronics, embedded programming and robotic movement from many fields. The EMG-Controlled Robotic Arm gives hands-on experience with sensors, signal handling, actuator coding, CAD modeling, 3D printing and system integration. The core idea is to turn muscle motion into a signal that drives the EMG-Controlled Robotic Arm. The EMG-Controlled Robotic Arm can. Be tested as a base, for studying how people and machines work together and how muscles can power robots.